

REVIEW 1: PROJECT PROPOSAL & DESIGN REVIEW DEFINITION

Course: Compiler Design Laboratory | Review 1 Evaluation (20 Marks)
Student Name: Lakshya Singh | Reg No.: 24BDS0054
Project Title: Sudarshan: An End-to-End Compiler and Virtual Machine
Repository: https://github.com/laksh1357/MyOWN_Compiler

1. Review 1 Marks Breakdown (20/20 Marks)

Criteria	What Evaluators Look For	Marks
1. Topic Selection	Compiler Design Relevance	3 / 3
2. Problem Statement	Clarity & Significance	3 / 3
3. Objectives	Clearly Defined Objectives	2 / 2
4. Technical Feasibility	Implementation Feasibility	4 / 4
5. Compiler Concepts	Appropriate Concepts Identified	3 / 3
6. System Architecture	Quality of Design & Data Flow	4 / 4
7. Innovation	Originality (Optimization/DCE)	2 / 2
8. Prototype	Initial Working Implementation	3 / 3
TOTAL MARKS		20 / 20

2. Panel Presentation Script & System Architecture

Problem Statement: Production compilers (GCC, Clang) are black boxes comprising millions of lines. Sudarshan provides a transparent, zero-dependency 5-phase compiler pipeline in Python 3.

Source Code (.mini)		
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Phase 1: Lexical Analysis (lexer.py)	â â â "	Token Stream (Line/Col info)
â		
Phase 2: Syntax Analysis (parser.py & ast_nodes.py)	â â â "	Abstract Syntax Tree (AST)
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Phase 3: Semantic Analysis (semantic.py)	â â â "	Scoped Symbol Table
â		
Phase 4: Intermediate Code Generator (tac.py)	â â â "	3-Address Code Quadruples
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Phase 4b: Code Optimization Pass (tac_opt.py)	â â â "	Constant Folding & DCE
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Phase 5: Virtual Machine Interpreter (interpreter.py)	â â â "	Code Execution & Output

3. Working Prototype Demonstration

Demonstrate live CLI in terminal during Review 1:

```
cd Sudarshan
./sudarshan examples/valid.mini --all
```